

Experiment 5 : Build a small rule-based expert system or logic inference engine for a diagnosis or advisory problem.

Objective : To design and implement a small rule-based expert system using logical inference techniques that provides diagnosis or advisory solutions based on predefined rules and user inputs.

Procedure

- 1.Open a Python IDE and create a new file named laptop_expert.py.
- 2.Define the laptop_expert() function with three input parameters: turns_on, internet_works, and is_slow.
- 3.Implement rule-based if-elif-else conditions to diagnose common laptop problems and provide appropriate advice.
- 4.Accept user inputs for the three questions and convert them to lowercase using .lower().
- 5.Call the laptop_expert() function with the user inputs and store the returned result.
- 6.Display the diagnosis or advisory message using the print() function.
- 7.Execute the program and test it with different input combinations to verify the output.

```
def laptop_expert(turns_on , internet_works, is_slow) :
    if turns_on == "no" :
        return "Check the charger and power cable."
    elif internet_works == "no" :
        return " Check Wi-Fi and restart the router."
    elif is_slow == "yes" :
        return "Close unused program and restart the laptop."
    else :
        return "No common problem was found in the laptop."
turns_on = input("Does the laptop turn on? yes/no: ").lower()
internet_works = input("Does the internet work? yes/no: ").lower()
is_slow = input("Is the laptop slow? yes/no: ").lower()

advice = laptop_expert(turns_on, internet_works, is_slow)
print(advice)
```

```
Does the laptop turn on? yes/no: no
Does the internet work? yes/no: no
Is the laptop slow? yes/no: yes
Check the charger and power cable.
```

Conclusion

The rule-based expert system was successfully implemented using Python. The program applies predefined if-elif-else rules to analyze user inputs and provides suitable diagnosis or advisory messages for common laptop problems.